

ASSIGNMENT 4

FILE TRANSFER USING TCP

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Aim: To Transfer a file from server to client using TCP socket programming

Sample Input Output

Client Side

Enter the path of file Desktop\File.txt

File Transferred.

Save the file in path: Desktop\newfile.txt

Server

Waiting for client

File to be transferred is Desktop\File.txt

File Transferred

Code:

Server

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
#define BUFFER_SIZE 1024

int main() {
    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);
    char buffer[BUFFER_SIZE] = {0};

    if ((server_fd = socket(AF_INET, SOCK_STREAM, 0)) == 0)
    {
        perror("Socket failed");
        exit(EXIT_FAILURE);
    }

    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(7777);

    if (bind(server_fd, (struct sockaddr *)&address, sizeof(address)) < 0)
    {
        perror("Bind failed");
        close(server_fd);
        exit(EXIT_FAILURE);
    }
    \
    if (listen(server_fd, 3) < 0)
    {
        perror("Listen failed");
        close(server_fd);
        exit(EXIT_FAILURE);
    }
```

```

printf("Waiting for client...\n");

if ((new_socket = accept(server_fd, (struct sockaddr *)&address, (socklen_t*)&addrlen)) <
0)
{
    perror("Accept failed");
    close(server_fd);
    exit(EXIT_FAILURE);
}

read(new_socket, buffer, BUFFER_SIZE);
printf("File to be transferred is %s\n", buffer);
FILE *file = fopen(buffer, "rb");
if (file == NULL)
{
    perror("File not found");
    close(new_socket);
    close(server_fd);
    exit(EXIT_FAILURE);
}

while (1)
{
    int bytes_read = fread(buffer, 1, BUFFER_SIZE, file);
    if (bytes_read > 0)
        send(new_socket, buffer, bytes_read, 0);

    if (bytes_read < BUFFER_SIZE)
    {
        if (feof(file))
            printf("File Transfer Completed.\n");
        if (ferror(file))
            perror("Error reading the file");
        break;
    }
}

fclose(file);
close(new_socket);
close(server_fd);
return 0;
}

```

Client

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
#define BUFFER_SIZE 1024

int main() {
    int sock = 0;
    struct sockaddr_in serv_addr;

```

```

char buffer[BUFFER_SIZE] = {0};

if ((sock = socket(AF_INET, SOCK_STREAM, 0)) < 0)
{
    printf("Socket creation error\n");
    return -1;
}

serv_addr.sin_family = AF_INET;
serv_addr.sin_port = htons(7777);

if (inet_pton(AF_INET, "127.0.0.1", &serv_addr.sin_addr) <= 0)
{
    printf("Invalid address/ Address not supported\n");
    return -1;
}

if (connect(sock, (struct sockaddr *)&serv_addr, sizeof(serv_addr)) < 0)
{
    printf("Connection Failed\n");
    return -1;
}

char file_path[BUFFER_SIZE];
printf("Enter the path of file: ");
scanf("%s", file_path);
send(sock, file_path, strlen(file_path), 0);
printf("File Transferred.\n");
printf("Save the file in path: ");
char save_path[BUFFER_SIZE];
scanf("%s", save_path);

FILE *file = fopen(save_path, "wb");
if (file == NULL)
{
    perror("Error creating file");
    close(sock);
    return -1;
}

while (1)
{
    int bytes_received = recv(sock, buffer, BUFFER_SIZE, 0);
    if (bytes_received > 0)
        fwrite(buffer, 1, bytes_received, file);

    if (bytes_received < BUFFER_SIZE)
    {
        if (feof(file))
            printf("File Transfer Completed.\n");
        if (ferror(file))
            perror("Error receiving the file");
        break;
    }
}

```

```
fclose(file);  
close(sock);  
return 0;  
}
```

Output:

Server Output:

```
nivethadhanakoti@Nivethas-MacBook-Pro A4 % gcc server.c -o server  
nivethadhanakoti@Nivethas-MacBook-Pro A4 % ./server  
Waiting for client...  
File to be transferred is Sample.txt  
File Transfer Completed.  
nivethadhanakoti@Nivethas-MacBook-Pro A4 %
```

Client Output:

```
nivethadhanakoti@Nivethas-MacBook-Pro A4 % gcc client.c -o client  
nivethadhanakoti@Nivethas-MacBook-Pro A4 % ./client  
Enter the path of file: Sample.txt  
File Transferred.  
Save the file in path: NewFile.txt
```

Learning Outcomes:

I have learnt to be proficient in file transfer using TCP.